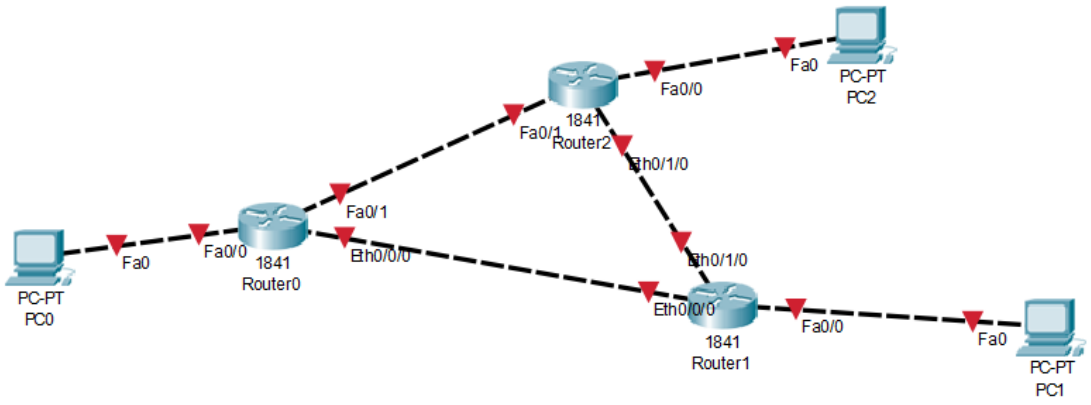
Start up packet tracer – this tutorial has been tested in version 7 and 8

1) Add three 1841 routers. Add in 2 modules with Ethernet ports (W1C-1ENET). You will need to turn the routers off and on again to do this.

2) Add three generic PCs.

3) Using "Copper crossover" wire the routers as shown in the diagram. (You will need options->preferences->Always show port labels in logical workspace).



You must make sure you have all the interfaces correctly wired and that you used the correct wire.

Assign the following. All have subnet mask 255.255.255.0. Switch them on. (If both ends of a connection are switched on they are green – if they are red they are off).

Device	Port	IP
PC 0	Fa0	192.168.0.1
PC 1	Fa0	192.168.1.1
PC 2	Fa0	192.168.2.1
Router 0	Fa0/0	192.168.0.2
Router 0	Fa0/1	192.168.3.1
Router 0	Eth0/0/0	192.168.4.1
Router 1	Fa0/0	192.168.1.2
Router 1	Eth0/0/0	192.168.4.2
Router 1	Eth0/1/0	192.168.5.1
Router 2	Fa0/0	192.168.2.2
Router 2	Fa0/1	192.168.3.2
Router 2	Eth0/1/0	192.168.5.2

For each PC you must also configure its GATEWAY address. This is the address of the router interface that PC is directly connected to. You need to work this out by looking at the table and the topology.

Now connect to PC0 and use the command prompt.

You should be able to type:

```
ping 192.168.0.2
```

If this is correct you should get four replies

```
Reply from 192.168.0.2: bytes=32 time=0ms TTL=255
Reply from 192.168.0.2: bytes=32 time=0ms TTL=255
Reply from 192.168.0.2: bytes=32 time=0ms TTL=255
Reply from 192.168.0.2: bytes=32 time=0ms TTL=255
```

```
tracert 192.168.0.2

Tracing route to 192.168.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.0.2

Trace complete.
```

If you have gone wrong then you will not get a ping response.
Also check PC1 by pinging 192.168.1.2 and PC2 by pinging 192.168.2.2

If you have a problem check the following.
1) Are all lights on the routers green. (If not are routers on and are interfaces on).
2) Are the IP addresses set correctly according to the table.
3) Do all PCs have a gateway address set.
4) Are all subnet masks set to 255.255.255.0 (same as a /24)

If you have done it right you will get ping responses from these.
However go back to PC0. If you ping 192.168.1.2 you will not get a response.

```
PC>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.0.2: Destination host unreachable.

Request timed out.

Reply from 192.168.0.2: Destination host unreachable.

Reply from 192.168.0.2: Destination host unreachable.

Ping statistics for 192.168.1.2:

    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Next you need to set up routing. The simplest way to do this is to add to RIP information about each network it must route for. On each router add into RIP the networks each router is attached to, from the set of networks. In this setting each router attaches to three networks from the set:
192.168.0.0, 192.168.1.0, 192.168.2.0, 192.168.3.0, 192.168.4.0, 192.168.5.0

Now on PC0 you should be able to do:

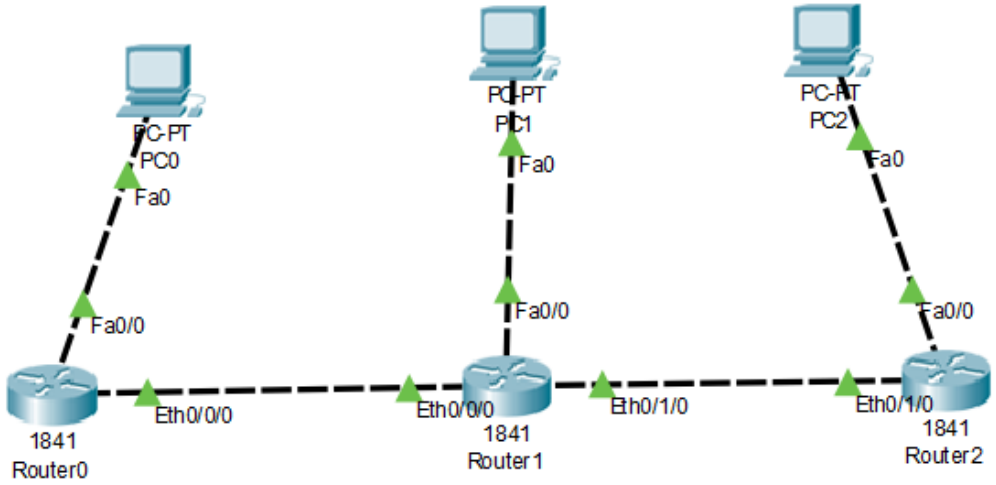
```
PC>ping 192.168.2.1

    Pinging 192.168.2.1 with 32 bytes of data:
```

```
    Reply from 192.168.2.1: bytes=32 time=0ms TTL=126
    Reply from 192.168.2.1: bytes=32 time=0ms TTL=126
    Reply from 192.168.2.1: bytes=32 time=0ms TTL=126
    Reply from 192.168.2.1: bytes=32 time=0ms TTL=126

    Ping statistics for 192.168.2.1:
        Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
tracert 192.168.2.1
    Tracing route to 192.168.2.1 over a maximum of 30 hops:
      1    1 ms      0 ms      0 ms      192.168.0.2
      2    0 ms      0 ms      0 ms      192.168.3.2
      3    0 ms      0 ms      0 ms      192.168.2.1
    Trace complete.
```

In the real lab you will need to be able to work out how to assign your own addresses to sub networks. Remember each segment of network separated by routers is a sub network. The previous example had SIX sub networks – one between each PC and the router (three) and one between each pair of routers (three more). To ensure that things are added into a routing table sometimes you must send a packet from a host so that it can be seen by the network.



Set up this topology with PC0 on address 190.0.0.1/16 PC1 on 190.1.0.1/16 and PC2 on 190.2.0.1/16 (the subnet mask is 255.255.0.0).
You need to be able to set this up with your own IP addresses without having the table provided in the first example. **If you are to get all marks in the router lab, you must be able to configure a topology like this one without any help.**